Midterm Exam

Course Title	Linear Algebra		Course Code	ELE0205	
Semester	First Semester		Academic Year	2025-2026	
Exam Date	December ,1st ,2025	Day	Monday	Time	1:30-2:20

اسمالطالب		الرقم الجامعي	ANSWERS	الشعبة		اسم المدرس	

GENERAL INSTRUCTIONS1

1. Students are not allowedtoleavetheexamroomduringthefirst30minutesoftheexam.
2. Late students are notpermittedtocompletetheirexamsafterthefirst30minutesunlessauthorizedbythe proctor (no studentslefttheexamsession),knowingthatnoadditionaltimewillbegranted.
3. Students must use ablueorblackpen.Pencilsarenotallowed.
4. Consumption of anyfoodorbeverageisprohibitedexceptinmedicalcases.
5. All electronic devices(e.g.,cellphones)areprohibited.
6. All cell phones mustbeturnedoffandplacedontheinvigilator'sdeskintheexamroom.
7. All notes and readingmaterialsmustbeplacedontheinvigilator'sdeskintheexamroom.
8. Students are not permittedtospeaktooraskquestionsoftheirclassmates.Ifastudenthasaquestion,heor she must raise his orherhandtobeassistedbytheproctor.
9. Write the full informationrequiredonthispagebeforecommencingyourexam.
10. Students are not allowedtoleavetheroomoncetheexamcommencesforanyreasonunlessitisabsolutely necessary.
11. Reported student cheatingcases,misbehavior,orotherstudentmisconductonexaminationsessionswillbe reported to the InvestigationsCommitteeasperinternalbylaws.

Ihave understood and abidewiththeexampolicy.Student'ssignature: _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
Part 1: 1-2,4,7	Multiple Choice	CILO 1	8 (26.67%)			
Part 1: 3,8-9	Multiple Choice	CILO 4	6 (20%)			
Part1: 10	Multiple Choice	CILO 8	2(6.67%)			
Part1: 5,6	Multiple Choice	CILO 3	4(13.33%)			
Part 2: 1	Essay Questions	CILO 9	4(13.33%)			
Part 2: 2	Essay Questions	CILO 8	6(20%)			
Total			30 (100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

جدول الاجابات

Question	1	2	3	4	5	6	7	8	9	10
Answer	✔	B	A	C	D	D	C	B	C	B

(20 Po

Part 1 : Multiple Choice Questions

1. If $A=[a_{ij}]$ is 2×2 matrix such that $a_{ij}=\begin{cases} i^2 & , \ i<j \\ i^2+j^2 & , \ i=j \\ 2 & , \ i>j \end{cases}$

then $A=$

A. $\begin{pmatrix} 2 & 1 \\ 2 & 8 \end{pmatrix}$

B. $\begin{pmatrix} 8 & 1 \\ 2 & 2 \end{pmatrix}$

C. $\begin{pmatrix} 2 & 1 \\ 8 & 2 \end{pmatrix}$

D. $\begin{pmatrix} 2 & 2 \\ 1 & 8 \end{pmatrix}$

2. Given that $C=\begin{pmatrix} 1 & 2 & -1 \\ 3 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ -1 & 1 \\ 2 & 4 \end{pmatrix}$ then $c_{22}=$

A. 21 B. 17 C. 1 D. 5

3. $\begin{vmatrix} 2 & 1 & -1 \\ 0 & 0 & 4 \\ 0 & 2 & 1 \end{vmatrix}=$

A. -16 B. 4 C. -4 D. 16

4. If $A=\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ then $A^6=$

A. $\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}$

B. $\begin{pmatrix} 16 & 16 \\ 16 & 16 \end{pmatrix}$

C. $\begin{pmatrix} 32 & 32 \\ 32 & 32 \end{pmatrix}$

D. $\begin{pmatrix} 64 & 64 \\ 64 & 64 \end{pmatrix}$

5. The system of the linear equations $\begin{cases} x+y=5 \\ k^2y-y=k+1 \end{cases}$ has no solution when $k=$

A. -1 B. 2 C. -2 D. 1

6. Given that $AX=B$ is a system of linear equations such that $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 2 & -1 & 1 \end{pmatrix} A = I$ and $B=\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$ then $X=$

A. $\begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$ B. $\begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix}$ C. $\begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$ D. $\begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}$

7. If A and B are two square matrices of the same size then $(A-B)(A+B)=$

A. A^2-B^2 C. $A^2+AB-BA-B^2$

B. $A^2-AB+BA-B^2$ D. B^2-A^2

8. If A and B are two 4×4 matrices such that $\det(A)=3$ and $\det(B)=5$ then $\det(2AB^T)=$

A. 30 B. 240 C. -30 D. -240

9. If A is a square matrix such that $A^3=\mathbf{0}$ then $(I+A)^{-1}=$

A. $A+A^2$ B. $I-A$ C. $I-A+A^2$ D. $I+A+A^2$

10. If $A\,\text{adj}(A)=\begin{pmatrix} 4 & x-3 \\ 0 & 2y \end{pmatrix}$ then $\det(A)+2x+y=$

A. 10 B. 12 C. 15 D. 17

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Question	1	2	3	4	5	6	7	8	9	10
Answer	✔	A	B	D	A	C	B	D	D	C

Part 1 : Multiple Choice Questions

(20 Points)

1) If $A=[a_{ij}]$ is 2×2 matrix such that $a_{ij}=\begin{cases} 2 & , \ i<j \\ i^2+j^2 & , \ i=j \\ i^2 & , \ i>j \end{cases}$

then $A=$

A) $\begin{pmatrix} 2 & 4 \\ 1 & 8 \end{pmatrix}$

B) $\begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix}$

C) $\begin{pmatrix} 8 & 1 \\ 4 & 2 \end{pmatrix}$

D) $\begin{pmatrix} 2 & 4 \\ 8 & 2 \end{pmatrix}$

2) Given that $C=\begin{pmatrix} 1 & 2 & -1 \\ 3 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ -1 & 1 \\ 2 & 4 \end{pmatrix}$ then $c_{32}=$

A) 21

B) 17

C) 1

D) 5

3) $\begin{vmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 4 \end{vmatrix} =$

A) -4

B) 16

C) -16

D) 4

4) If $A=\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ then $A^7=$

A) $\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}$

B) $\begin{pmatrix} 16 & 16 \\ 16 & 16 \end{pmatrix}$

C) $\begin{pmatrix} 32 & 32 \\ 32 & 32 \end{pmatrix}$

D) $\begin{pmatrix} 64 & 64 \\ 64 & 64 \end{pmatrix}$

Name: _____

ID: A

5) The system of the linear equations $\begin{cases} x+y=5 \\ k^2y-y=k+1 \end{cases}$ has infinite solution when $k=$

A) -1 B) 2 C) -2 D) 1

6) Given that $AX=B$ is a system of linear equations such that $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 2 & -1 & 1 \end{pmatrix} A=I$ and $B=\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ then $X=$

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7) If A and B are two square matrices of the same size then $(A+B)(A-B)=$

A) A^2-B^2 C) $A^2+AB-BA-B^2$

B) $A^2-AB+BA-B^2$ D) B^2-A^2

8) If A and B are two 4×4 matrices such that $\det(A)=3$ and $\det(B)=-5$ then $\det(2AB^T)=$

A) 30 B) -30 C) 240 D) -240

9) If A is a square matrix such that $A^3=\mathbf{0}$ then $(I-A)^{-1}=$

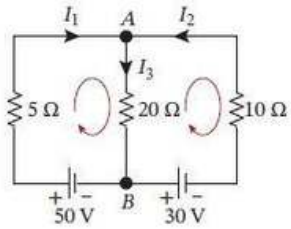
A) $A+A^2$ B) $I-A$ C) $I-A+A^2$ D) $I+A+A^2$

10) If $A\,\mathrm{adj}(A)=\begin{pmatrix} 4 & x-3 \\ 0 & 2y \end{pmatrix}$ then $\det(A)+3x+y=$

A) 10 B) 12 C) 15 D) 17

1. Applying Kirchoff Laws on the electrical circuit inthefigure,givesthegivensystemof linear equations

$$I1+I_2-i_3=0$$
$$I1 \quad +4I_3=10$$
$$I\,2+2I_3=-3$$



This picture, from the textbook

Using Cramer rule, find I_3 only. (4Points)

$$I_3=\frac{\begin{vmatrix}1&1&0\\1&0&10\\0&1&-3\end{vmatrix}}{\begin{vmatrix}1&1&-1\\1&0&4\\0&1&2\end{vmatrix}}$$

1 point

$$=\frac{-7}{-7}$$

2 points

$$\therefore I_3=1A$$

1 point

2. Given that $\begin{vmatrix}a&b&c\\d&e&f\\g&h&i\end{vmatrix}=7$, then (6 Points)

a. $\begin{vmatrix}a&b&c\\2d&2e&2f\\g&h&i\end{vmatrix}=$ ……14…… 2 points

b. $\begin{vmatrix}a&c&b\\d&f&e\\g&i&h\end{vmatrix}=$……−7…… 2 points

c. $\begin{vmatrix}a&b&c\\5a+d&5b+e&5c+f\\g&h&i\end{vmatrix}=$……7…… 2 points